A OMITTED DATASET DETAILS FOR EXPERIMENTS

Table 2: Datasets

Dataset	# sample	M	K	R	E	b	Model
MNIST	60,000	300	5	30	10	10	CNN
FashionM	60,000	300	5	50	10	10	CNN
CIFAR10	60,000	600	5	50	10	10	VGG16
CIFAR100	60,000	600	10	50	10	10	VGG16
FEMNIST	811,586	3,556	5	350	20	10	CNN
Shakes	3,678,451	660	20	30	100	60	LSTM

B OMITTED PROOFS FOR UNLEARNING ANALYSIS

B.1 Proof of Claim 1

Proof of Claim 1. We concentrate on the mini-batch sampling measure at client k_u, since the unlearning operation only affects the mini-batches sampled by client k_u. Specifically, we denote by $\xi_{(N,b)}$ the probability measure of k_u's sampling a mini-batch of size b from $\mathcal{D}_{k_u}$ and by $\xi^{\mathtt{dul}}_{(N,b)} := \mathtt{dul}\#\xi_{(N,b)}$. Similarly, we denote by $\xi_{(N-1,b)}$ the probability measure for sampling a b size mini-batch from $\mathcal{D}_{k_u} \setminus \{X_u\}$. Let $B \in (\mathcal{D}_{k_u})^b$ be any possible mini-batch. We show $\xi^{\mathtt{dul}}_{(N,b)}(B) = \xi_{(N-1,b)}(B)$ for each local iteration by two cases.

Case 1. If the verification fails, i.e., a re-computation is triggered, the algorithm will re-sample $B \sim \xi_{(N-1,b)}$. Thus, $\xi^{\mathtt{dul}}_{(N,b)}(B) = \xi_{(N-1,b)}(B)$ holds trivially for this case.

Case 2. If the verification succeeds. Then the deleted data X_u is not contained in B. The measure $\xi^{\mathtt{dul}}_{(N,b)}$ is therefore just the probability under the original sampling measure $\xi_{(N,b)}$ conditioned on the event that $X_u \notin B$. Therefore,

$$\xi^{\mathtt{dul}}_{(N,b)}(B) = \xi_{(N,b)}(B|X_u \notin B) = \frac{\xi_{(N,b)}(B \cap \{X_u \notin B\})}{\xi_{(N,b)}(\{X_u \notin B\})},$$

where, by direct computation, $\xi_{(N,b)}(\{X_u \notin B\}) = 1 - \xi_{(N,b)}(\{X_u \in B\}) = 1 - \frac{\binom{N-1}{b-1}}{\binom{N}{b}} = 1 - \frac{b}{N}$. To calculate the numerator, we consider two possible choices of B. First, if $X_u \in B$, then $\xi_{(N,b)}(B \cap \{X_u \notin B\}) = 0$, which gives us that $\xi^{\mathtt{dul}}_{(N,b)}(B) = 0 = \xi_{(N-1,b)}(B)$. For the choice that $X_u \notin B$,

$$\begin{aligned}
\xi^{\mathtt{dul}}_{(N,b)}(B) &= \frac{\xi_{(N-1,b)}(B)}{\xi_{(N-1,b)}(\{X_u \notin B\})} = \frac{1/\binom{N}{b}}{1-\frac{b}{N}} \\
&= \frac{N}{N-b}\frac{(N-b)!b!}{N!} = \frac{(N-b-1)!b!}{(N-1)!} \\
&= \frac{1}{\binom{N-1}{b}} = \xi_{(N-1,b)}(B).
\end{aligned}$$

Therefore, $\xi^{\mathtt{dul}}_{(N,b)}(B) = \xi_{(N-1,b)}(B)$ also holds for Case 2. □

B.2 Proof of Claim 3

Proof of Claim 3.

$$\begin{aligned}
&\mathbb{P}_{(\mathcal{H},\mathcal{H}')}(\mathcal{H} \neq \mathcal{H}') \\
&= \mathbb{P}_{(\mathcal{H},\mathcal{H}')}(\exists r \in [R] \text{ s.t. } (\mathcal{P}[r], \mathcal{B}[r]) \neq (\mathcal{P}'^{[r]}, \mathcal{B}'^{[r]})) \\
&= \mathbb{P}_{\mathcal{H}}(\exists r \in [R], i \in [E] \text{ s.t. } X_u \in \mathcal{B}^{[r,i]}_{k_u}) \leq \frac{TKb}{MN} = \rho_S.
\end{aligned}$$

□

C OMITTED PROOFS FOR CONVERGENCE ANALYSIS

C.1 Proof of Lemma 2

Proof of Lemma 2. Our algorithm involves two sources of stochasticity. The first one arises from the stochastic gradient, while the second one stems from the client sampling. To differentiate them, we adopt the notation $\mathbb{E}_{\mathcal{B}^{(t)}} := \mathbb{E}_{\mathcal{B}_1^{(t)},\ldots,\mathcal{B}_M^{(t)}}$ to indicate the expectation with respect to the randomness of mini-batch sampling in iteration t, and use $\mathbb{E}_{\mathcal{P}^{(t)}}$ to eliminate the randomness of client sampling in iteration t. Let $\mathbb{E}_t$ be the expectation with respect to all of the randomness until time t, then $\mathbb{E}_t = \mathbb{E}_{\mathcal{B}^{(t)}}\mathbb{E}_{\mathcal{P}^{(t)}}\mathbb{E}_{t-1}$. We drop the time indicator t and simply use $\mathbb{E}$ to denote the expectation with respect to all the relevant randomness when it is clear from the context.

Using the update rule $\theta_k^{(t)} \leftarrow \theta_k^{(t-1)} - \eta \cdot \tilde{g}_k^{(t)}$ and the L-smoothness of the loss function f (Assumption 1), we derive the following inequality:

$$F(\theta^{(t)}) - F(\theta^{(t-1)}) \leq -\eta\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)}\rangle + \frac{\eta^2 L}{2}\|\tilde{g}^{(t)}\|_2^2.$$

Taking expectation on both sides, we obtain

$$\mathbb{E}[F(\theta^{(t)}) - F(\theta^{(t-1)})] \leq -\eta\mathbb{E}[\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)}\rangle] + \frac{\eta^2 L}{2}\mathbb{E}[\|\tilde{g}^{(t)}\|_2^2].$$

Averaging over all iterations $t = 1, 2, \ldots, T$, we get

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[F(\theta^{(t)}) - F(\theta^{(t-1)})] \leq -\frac{\eta}{T}\sum_{t=1}^{T}\mathbb{E}[\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)}\rangle] + \frac{\eta^2 L}{2T}\sum_{t=1}^{T}\mathbb{E}[\|\tilde{g}^{(t)}\|_2^2]. \tag{11}$$

To further bound the terms on the right hand side of the above inequality, we need the following results.

Claim 4. The expected inner product between the stochastic and the full-batch gradients in iteration t can be bounded as follows:

$$-\eta\mathbb{E}[\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)}\rangle] \leq -\frac{\eta}{2}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2}\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta L^2}{2M}\sum_{k=1}^{M}\mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2].$$

Claim 5. The averaged distance of local models from their virtual average during the learning process holds for

$$\frac{1}{TM}\sum_{t=1}^{T}\sum_{k=1}^{M}\mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2] \leq \frac{E(K+1)\eta^2 G^2}{Kb} + \frac{2\eta^2\lambda E(E-1)}{T}\sum_{t=1}^{T}\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2]$$

Claim 6. The expected squared norm of the averaged stochastic gradient of iteration t can be bounded as follows:

$$\mathbb{E}[\|\tilde{g}^{(t)}\|_2^2] \leq \frac{G^2}{Kb} + \lambda\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2]$$

We continue the proof by utilizing Claim 4, claim 5 and claim 6 to further bound (11) as follows:

$$\begin{aligned}
&\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[F(\theta^{(t)}) - F(\theta^{(t-1)})] \\
&\leq \frac{1}{T}\sum_{t=1}^{T}(-\eta\mathbb{E}[\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)}\rangle]) + \frac{1}{T}\sum_{t=1}^{T}\frac{\eta^2 L}{2}\mathbb{E}[\|\tilde{g}^{(t)}\|_2^2] \\
&\leq \frac{1}{T}\sum_{t=1}^{T}\Big(-\frac{\eta}{2}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2}\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta L^2}{2M}\sum_{k=1}^{M}\mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2]\Big) + \frac{1}{T}\sum_{t=1}^{T}(\frac{\eta^2 L}{2}(\frac{G^2}{Kb} + \lambda\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2])) \\
&= -\frac{\eta}{2T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] + (-\frac{\eta}{2} + \eta^3 L^2\lambda E(E-1) + \frac{\eta^2\lambda L}{2})\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta^3 L^2 G^2 E(K+1)}{2Kb} + \frac{\eta^2 L G^2}{2Kb} \\
&\leq -\frac{\eta}{2T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] + \frac{\eta^3 L^2 G^2 E(K+1)}{2Kb} + \frac{\eta^2 L G^2}{2Kb},
\end{aligned}$$

where the last inequality is due to the condition that $-\frac{\eta}{2} + \eta^3 L^2\lambda E(E-1) + \frac{\eta^2\lambda L}{2} < 0$.

By rearranging, we get

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{2(F(\theta^{(0)}) - F^*)}{\eta T} + \frac{\eta^2 L^2 G^2 E(K+1)}{Kb} + \frac{\eta L G^2}{Kb}.$$

Finally, by noting that $K = \frac{\rho_C E M}{T}$ and $b = \frac{\rho_S N}{\rho_C E}$, we can conclude the proof. □

C.2 Proof of Claim 4

Proof of Claim 4.

$$
\begin{aligned}
&- \eta \mathbb{E}_t[\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)} \rangle] \\
&= -\eta \mathbb{E}_{t-1} \mathbb{E}_{\mathcal{B}^{(t)}} [\mathbb{E}_{\mathcal{P}^{(t)}} [\langle \nabla F(\theta^{(t-1)}), \tilde{g}^{(t)} \rangle]] \\
&= -\eta \mathbb{E}_{t-1} \mathbb{E}_{\mathcal{B}^{(t)}} [\mathbb{E}_{\mathcal{P}^{(t)}} [\langle \nabla F(\theta^{(t-1)}), \frac{1}{K} \sum_{k \in \mathcal{P}^{(t)}} \tilde{g}_k^{(t)} \rangle]] \\
&\overset{①}{=} -\eta \mathbb{E}_{t-1} \mathbb{E}_{\mathcal{P}^{(t)}} [\mathbb{E}_{\mathcal{B}^{(t)}} [\langle \nabla F(\theta^{(t-1)}), \frac{1}{K} \sum_{k \in \mathcal{P}^{(t)}} \tilde{g}_k^{(t)} \rangle]] \\
&= -\eta \mathbb{E}_{t-1} [\langle \nabla F(\theta^{(t-1)}), \mathbb{E}_{\mathcal{P}^{(t)}} [\frac{1}{K} \sum_{k \in \mathcal{P}^{(t)}} \mathbb{E}_{\mathcal{B}^{(t)}} [\tilde{g}_k^{(t)}]] \rangle] \\
&= -\eta \mathbb{E}_{t-1} [\langle \nabla F(\theta^{(t-1)}), \frac{1}{K} \mathbb{E}_{\mathcal{P}^{(t)}} [\sum_{k \in \mathcal{P}^{(t)}} g_k^{(t)}] \rangle] \\
&= -\eta \mathbb{E}_{t-1} [\langle \nabla F(\theta^{(t-1)}), \frac{1}{K} [K \sum_{k=1}^{M} \frac{1}{M} g_k^{(t)}] \rangle] \\
&\overset{②}{=} -\frac{\eta}{2} \mathbb{E}_{t-1} [\|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2} \mathbb{E}_{t-1} [\|\frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta}{2} \mathbb{E}_{t-1} [\|\nabla F(\theta^{(t-1)}) - \frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2] \\
&= -\frac{\eta}{2} \mathbb{E}_{t-1} \|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2} \mathbb{E}_{t-1} [\|\frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta}{2} \mathbb{E}_{t-1} [\|\frac{1}{M} \sum_{k=1}^{M} (\nabla F_k(\theta^{(t-1)}) - g_k^{(t)})\|_2^2] \\
&\overset{③}{\leq} -\frac{\eta}{2} \mathbb{E}_{t-1} [\|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2} \mathbb{E}_{t-1} [\|\frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta}{2} \frac{1}{M} \sum_{k=1}^{M} \mathbb{E}_{t-1} [\|\nabla F_k(\theta^{(t-1)}) - \nabla F_k(\theta_k^{(t-1)})\|_2^2] \\
&\overset{④}{\leq} -\frac{\eta}{2} \mathbb{E}_{t-1} [\|\nabla F(\theta^{(t-1)})\|_2^2] - \frac{\eta}{2} \mathbb{E}_{t-1} [\|\frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2] + \frac{\eta}{2} \frac{L^2}{M} \sum_{k=1}^{M} \mathbb{E}_{t-1} [\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2],
\end{aligned}
$$

where ① is due to the fact that the randomness of $\mathcal{B}^{(t)}$ and $\mathcal{P}^{(t)}$ are independent, ② is due to the equation $2\langle a, b \rangle = \|a\|_2^2 + \|b\|_2^2 - \|a - b\|_2^2$, ③ holds because of convexity of $\|\cdot\|_2$, and ④ follows from L-smoothness of loss function f (Assumption 1). □

C.3 Proof of Claim 5

Proof of Claim 5. Define $t_c \coloneqq \lfloor \frac{t-1}{E} \rfloor E$, which denotes the last iteration before entering the current round of iteration t. The local model at client k in any iteration $t > t_c$ can be expressed as

$$
\theta_k^{(t)} = \theta_k^{(t-1)} - \eta \tilde{g}_k^{(t)} = \theta_k^{(t-2)} - [\eta \tilde{g}_k^{(t-2)} + \eta \tilde{g}_k^{t-1}] = \ldots = \theta_k^{t_c} - \sum_{\tau=t_c+1}^{t} \eta \tilde{g}_k^{(\tau)}. \tag{12}
$$

From (12), we compute the virtual average model in t-th iteration as follows:

$$
\theta^{(t)} = \theta^{(t_c)} - \frac{\eta}{K} \sum_{k \in \mathcal{P}^{(t)}} \sum_{\tau=t_c+1}^{t} \tilde{g}_k^{(\tau)}.
$$

We express t as $t = sE + r$, where $s \in \{0, 1, \ldots, \lfloor \frac{T-1}{E} \rfloor\}$ denotes the indices of communication round and $r \in [E]$ denotes the indices of local updates. Observe that for $t_c < t \leq t_c + E$, $\mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2$ does not depend on time steps $t \leq t_c$ for all client $k \in [M]$. Thus, we obtain the following for all iterations $1 \leq t \leq T$:

$$
\frac{1}{TM} \sum_{t=1}^{T} \sum_{k=1}^{M} \mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2] = \frac{1}{TM} \sum_{s=0}^{\lfloor \frac{T-1}{E} \rfloor} \sum_{r=1}^{E} \sum_{k=1}^{M} \mathbb{E}[\|\theta^{(sE+r-1)} - \theta_k^{(sE+r-1)}\|_2^2].
$$

We first bound the term $\mathbb{E}[\|\theta^{(sE+r-1)} - \theta_k^{(sE+r-1)}\|_2^2]$ for any fixed tuple of (s, r, k).

$\mathbb{E}[\|\theta^{(sE+r-1)} - \theta_k^{(sE+r-1)}\|_2^2]$

$$
\begin{aligned}
&= \mathbb{E}[\|\theta^{(sE)} - \frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)} - \theta^{(sE)} + \sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)}\|_2^2] \\
&= \mathbb{E}[\|\sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)} - \frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)}\|_2^2] \\
&\overset{①}{\leq} 2\mathbb{E}[\|\sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)}\|_2^2 + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)}\|_2^2] \\
&\overset{②}{=} 2\mathbb{E}[\|\sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)} - \mathbb{E}[\sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)}]\|_2^2 + \|\mathbb{E}[\sum_{i=1}^{r-1}\eta\tilde{g}_k^{(sE+i)}]\|_2^2 + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)} - \mathbb{E}[\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)}]\|_2^2 + \|\mathbb{E}[\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta\tilde{g}_l^{(sE+i)}]\|_2^2] \\
&= 2\mathbb{E}\Bigg[\|\sum_{i=1}^{r-1}\eta(\tilde{g}_k^{(sE+i)} - g_k^{(sE+i)})\|_2^2 + \|\sum_{i=1}^{r-1}\eta g_k^{(sE+i)}\|_2^2 + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta(\tilde{g}_l^{(sE+i)} - g_l^{(sE+i)})\|_2^2 + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta g_l^{(sE+i)}\|_2^2\Bigg] \\
&= 2\mathbb{E}\Bigg[\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_k^{(sE+i)} - g_k^{(sE+i)}\|_2^2 + \sum_{u\neq v}\langle\eta\tilde{g}_k^{(u)} - \eta g_k^{(u)}, \eta\tilde{g}_k^{(v)} - \eta g_k^{(v)}\rangle + \|\sum_{i=1}^{r-1}\eta g_k^{(sE+i)}\|_2^2 \\
&\quad + \frac{1}{K^2}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_l^{(sE+i)} - g_l^{(sE+i)}\|_2^2 + \frac{1}{K^2}\sum_{u\neq v, y\neq z}\langle\eta\tilde{g}_y^{(u)} - \eta g_y^{(u)}, \eta\tilde{g}_z^{(v)} - \eta g_z^{(v)}\rangle + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta g_l^{(sE+i)}\|_2^2\Bigg] \\
&\overset{③}{=} 2\mathbb{E}\Bigg[\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_k^{(sE+i)} - g_k^{(sE+i)}\|_2^2 + \|\sum_{i=1}^{r-1}\eta g_k^{(sE+i)}\|_2^2 + \frac{1}{K^2}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_l^{(sE+i)} - g_l^{(sE+i)}\|_2^2 + \|\frac{1}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta g_l^{(sE+i)}\|_2^2\Bigg] \\
&\overset{④}{\leq} 2\mathbb{E}[\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_k^{(sE+i)} - g_k^{(sE+i)}\|_2^2 + (r-1)\eta^2\sum_{i=1}^{r-1}\|g_k^{(sE+i)}\|_2^2 + \frac{1}{K^2}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\eta^2\|\tilde{g}_l^{(sE+i)} - g_l^{(sE+i)}\|_2^2 + \frac{(r-1)\eta^2}{K}\sum_{l\in\mathcal{P}^{(t)}}\sum_{i=1}^{r-1}\|g_l^{(sE+i)}\|_2^2] \\
&= 2\sum_{i=1}^{r-1}\eta^2\mathbb{E}[\|\tilde{g}_k^{(sE+i)} - g_k^{(sE+i)}\|_2^2] + 2(r-1)\eta^2\sum_{i=1}^{r-1}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \frac{2\eta^2}{KM}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|\tilde{g}_l^{(sE+i)} - g_l^{(sE+i)}\|_2^2] + \frac{2(r-1)\eta^2}{M}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2] \\
&\overset{⑤}{\leq} 2(r-1)\eta^2\frac{G^2}{b} + 2(r-1)\eta^2\sum_{i=1}^{r-1}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \frac{2(r-1)\eta^2}{K}\frac{G^2}{b} + \frac{2(r-1)\eta^2}{M}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2] \\
&= \frac{2(K+1)(r-1)\eta^2G^2}{Kb} + 2(r-1)\eta^2\sum_{i=1}^{r-1}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \frac{2(r-1)\eta^2}{M}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2],
\end{aligned}
$$

where ① is due to the triangle inequality $\|a+b\|_2 \leq \|a\|_2 + \|b\|_2$ and the Cauchy—Schwarz inequality, ② follows from $\mathbb{E}[w^2] = \mathbb{E}[(w - \mathbb{E}[w])^2] + (\mathbb{E}[w])^2$, ③ is due to the independent mini-batch sampling and unbiased gradient estimation, ④ uses Cauchy—Schwarz inequality again, and ⑤ is due to bounded local variance assumption of local stochastic gradients (Assmption 2).

Next, we let (s, r, k) vary and bound $\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{r=1}^{E}\sum_{k=1}^{M}\mathbb{E}[\|\theta^{(sE+r-1)} - \theta_k^{(sE+r-1)}\|_2^2]$.

$$
\begin{aligned}
&\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{r=1}^{E}\sum_{k=1}^{M}\mathbb{E}[\|\theta^{(sE+r-1)} - \theta_k^{(sE+r-1)}\|_2^2] \\
&= \sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{r=1}^{E}\sum_{k=1}^{M}\left(\frac{2(K+1)(r-1)\eta^2G^2}{Kb} + 2(r-1)\eta^2\sum_{i=1}^{r-1}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \frac{2(r-1)\eta^2}{M}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2]\right) \\
&\leq \frac{TME(K+1)\eta^2G^2}{Kb} + 2\eta^2\sum_{r=1}^{E}(r-1)\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{k=1}^{M}\sum_{i=1}^{r-1}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \frac{2\eta^2}{M}\sum_{r=1}^{E}(r-1)\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{k=1}^{M}\sum_{i=1}^{r-1}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2] \\
&\leq \frac{TME(K+1)\eta^2G^2}{Kb} + \eta^2E(E-1)\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{i=1}^{E}\sum_{k=1}^{M}\mathbb{E}[\|g_k^{(sE+i)}\|_2^2] + \eta^2E(E-1)\sum_{s=0}^{\lfloor\frac{T-1}{E}\rfloor}\sum_{i=1}^{E}\sum_{l=1}^{M}\mathbb{E}[\|g_l^{(sE+i)}\|_2^2]
\end{aligned}
$$

$$= \frac{TME(K+1)\eta^2 G^2}{Kb} + 2\eta^2 E(E-1) \sum_{s=0}^{\lfloor \frac{T-1}{E} \rfloor} \sum_{i=1}^{E} \sum_{k=1}^{M} \mathbb{E}[\|g_k^{(sE+i)}\|_2^2]$$
$$= \frac{TME(K+1)\eta^2 G^2}{Kb} + 2\eta^2 E(E-1) \sum_{t=1}^{T} \sum_{k=1}^{M} \mathbb{E}[\|g_k^{(t)}\|_2^2]$$

Finally, we get

$$\frac{1}{TM} \sum_{t=1}^{T} \sum_{k=1}^{M} \mathbb{E}[\|\theta^{(t-1)} - \theta_k^{(t-1)}\|_2^2] \leq \frac{E(K+1)\eta^2 G^2}{Kb} + \frac{2\eta^2 E(E-1)}{TM} \sum_{t=1}^{T} \sum_{k=1}^{M} \mathbb{E}[\|g_k^{(t)}\|_2^2]$$
$$\leq \frac{E(K+1)\eta^2 G^2}{Kb} + \frac{2\eta^2 \lambda E(E-1)}{T} \sum_{t=1}^{T} \mathbb{E}[\|\frac{1}{M} \sum_{k=1}^{M} g_k^{(t)}\|_2^2],$$

where the last inequality is due to our bounded heterogeneity assumption (Assumption 3). □

C.4 Proof of Claim 6

Proof of Claim 6.

$$\begin{aligned}
&\mathbb{E}[\|\tilde{g}^{(t)}\|_2^2] \\
&= \mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}\mathbb{E}_{\mathcal{B}^{(t)}}[\|\tilde{g}^{(t)}\|_2^2] \\
&= \mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}\mathbb{E}_{\mathcal{B}^{(t)}}[\|\tilde{g}^{(t)} - \mathbb{E}_{\mathcal{B}^{(t)}}[\tilde{g}^{(t)}]\|_2^2 + \|\mathbb{E}_{\mathcal{B}^{(t)}}[\tilde{g}^{(t)}]\|_2^2] \\
&= \mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}\mathbb{E}_{\mathcal{B}^{(t)}}[\|(\frac{1}{K}\sum_{k\in\mathcal{P}^{(t)}} \tilde{g}_k^{(t)}) - (\frac{1}{K}\sum_{k\in\mathcal{P}^{(t)}} g_k^{(t)})\|_2^2] + \mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}[\|\frac{1}{K}\sum_{k\in\mathcal{P}^{(t)}} g_k^{(t)}\|_2^2] \\
&\leq \frac{1}{K^2}\mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}\mathbb{E}_{\mathcal{B}^{(t)}}[\sum_{k\in\mathcal{P}^{(t)}} \|\tilde{g}_k^{(t)} - g_k^{(t)}\|_2^2] + \frac{1}{K}\mathbb{E}_{t-1}\mathbb{E}_{\mathcal{P}^{(t)}}[\sum_{k\in\mathcal{P}^{(t)}} \|g_k^{(t)}\|_2^2] \\
&\leq \frac{1}{K^2}\mathbb{E}_{t-1}[K\sum_{k=1}^{M}\frac{1}{M}\frac{G^2}{b}] + \frac{1}{K}\mathbb{E}_{t-1}[K\sum_{k=1}^{M}\frac{1}{M}\|g_k^{(t)}\|_2^2] \\
&\leq \frac{G^2}{Kb} + \lambda\mathbb{E}_{t-1}[\|\frac{1}{M}\sum_{k=1}^{M} g_k^{(t)}\|_2^2],
\end{aligned}$$

where in the last inequality we use the bounded heterogeneity assumption (Assumption 3). □

C.5 Proof of Theorem 2

Proof of Theorem 2. The choice of $\eta = \frac{1}{L\sqrt{\Gamma}T}$ comes by balancing the first and the last term in (9). With this choice of η, condition (7) becomes $\frac{E(E-1)}{T^2} < \frac{\Gamma}{2\lambda} - \frac{\sqrt{\Gamma}}{2T}$. By letting $T > \frac{2\lambda}{\sqrt{\Gamma}}$, it suffices to have $\frac{E}{T} < \frac{1}{2}\sqrt{\frac{\Gamma}{\lambda}}$ hold. Next we bound the mean-squared loss gradient norm. By the choice of η, we can calculate the sum of the first and the last term in (9) as

$$\frac{2(F(\theta^{(0)}) - F^*)}{\eta T} + \frac{\eta L G^2 T}{\rho_S MN} = \frac{3G\sqrt{L(F(\theta^{(0)}) - F^*)}}{\sqrt{\rho_S MN}}.$$

By taking the choice of η, we can calculate the the second term in (9) as

$$\frac{\eta^2 L^2 G^2 E(\rho_C EM + T)}{\rho_S MN} = L(F(\theta^{(0)}) - F^*)\frac{E}{T}(\frac{\rho_C ME}{T} + 1).$$

Combining these results, we obtain the desired bound. □

C.6 Proof of Corollary 1

Proof of Corollary 1. We start with the case of $E = T^{1-\alpha}$. Firstly, in order to have condition (7) hold, we require $T \geq \frac{2\lambda}{\sqrt{\Gamma}}$ and $\frac{E}{T} < \frac{1}{2}\sqrt{\frac{\Gamma}{\lambda}}$, which gives $T \geq \max\left\{\frac{2\lambda}{\sqrt{\Gamma}}, \left(2\sqrt{\frac{\lambda}{\Gamma}}\right)^{\frac{1}{\alpha}}\right\}$. Taking $E = T^{1-\alpha}$ into (10), we obtain

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{3\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} + \frac{L(F(\theta^{(0)})-F^*)}{T^\alpha}(\frac{\rho_C M}{T^\alpha}+1). \tag{13}$$

Furthermore, when T is large enough such that $T > (\rho_C M)^{\frac{1}{\alpha}}$, we have $\frac{\rho_C M}{T^\alpha} < 1$, which leads to the following

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{3G\sqrt{L(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} + \frac{2L(F(\theta^{(0)})-F^*)}{T^\alpha}. \tag{14}$$

Note that we have required $T \geq \left(2\sqrt{\frac{\lambda}{\Gamma}}\right)^{\frac{1}{\alpha}}$, we have

$$\frac{2L(F(\theta^{(0)})-F^*)}{T^\alpha} \leq \frac{L(F(\theta^{(0)})-F^*)}{\sqrt{\lambda}}\sqrt{\frac{G^2}{L(F(\theta^{(0)})-F^*)\rho_S MN}} \tag{15}$$

$$= \frac{G\sqrt{L(F(\theta^{(0)})-F^*)}}{\sqrt{\lambda\rho_S MN}} \tag{16}$$

$$\leq \frac{G\sqrt{L(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}}, \tag{17}$$

where the last inequality is due to the fact that $\lambda \geq 1$. Therefore, when $T \geq \max\left\{\frac{2\lambda}{\sqrt{\Gamma}}, \left(2\sqrt{\frac{\lambda}{\Gamma}}\right)^{\frac{1}{\alpha}}, (\rho_C M)^{\frac{1}{\alpha}}\right\}$, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{4G\sqrt{L(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} = O\left(\frac{1}{\sqrt{\rho_S MN}}\right). \tag{18}$$

Next, we move onto the case of $E = c \cdot T$ for some constant $c < \frac{1}{2}\sqrt{\frac{\Gamma}{\lambda}}$ and $T \geq \frac{2\lambda}{\sqrt{\Gamma}}$. By direct calculation, we have

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{3\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} + L(F(\theta^{(0)})-F^*)\frac{E}{T}(\frac{\rho_C ME}{T}+1) \tag{19}$$

$$\leq \frac{3\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} + \frac{L(F(\theta^{(0)})-F^*)\rho_C M\Gamma}{4\lambda} + \frac{L(F(\theta^{(0)})-F^*)}{2}\sqrt{\frac{\Gamma}{\lambda}} \tag{20}$$

$$= \frac{3\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} + \frac{\rho_C G^2}{4\lambda\rho_S N} + \frac{\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{2\sqrt{\lambda\rho_S MN}} \tag{21}$$

$$\leq \frac{7\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{2\sqrt{\rho_S MN}} + \frac{\rho_C G^2}{4\lambda\rho_S N}, \tag{22}$$

where the last inequality follows from the fact of $\lambda \geq 1$. When $M < \frac{4\lambda^2 L(F(\theta^{(0)})-F^*)\rho_S}{G^2\rho_C^2}N$, we obtain that $\frac{\rho_C G^2}{4\lambda\rho_S N} < \frac{\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{2\sqrt{\rho_S MN}}$, which implies that

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\|\nabla F(\theta^{(t-1)})\|_2^2] \leq \frac{4\sqrt{LG^2(F(\theta^{(0)})-F^*)}}{\sqrt{\rho_S MN}} = O\left(\frac{1}{\sqrt{\rho_S MN}}\right). \tag{23}$$

□